THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — MATHS · EASY LEVEL CHAPTER TEST SET 2

Chapter 14: **Probability** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Easy level: direct, single-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. A bag contains 4 red, 6 green balls. One ball is drawn at random. The probability that it is green is: 1

(a) $\frac{7}{10}$ **(b)** $\frac{2}{5}$ **(c)** $\frac{1}{2}$ **(d)** $\frac{3}{5}$

2. 12 cards numbered 1 to 12 are placed in a box, mixed well and one is drawn at random. The probability that the number drawn is a prime number is: 1

(a) $\frac{5}{12}$ **(b)** $\frac{1}{2}$ **(c)** $\frac{1}{3}$ **(d)** $\frac{7}{12}$

3. The probability of an event E is $\frac{3}{7}$. The probability of the event 'not E' is: 1

(a) $\frac{1}{14}$ **(b)** $\frac{10}{7}$ **(c)** $\frac{3}{7}$ **(d)** $\frac{4}{7}$

4. The probability of an impossible event is: 1

(a) -1 **(b)** 1 **(c)** $\frac{1}{2}$ **(d)** 0

5. A die is thrown once. The probability of getting a number less than 3 is: 1

(a) $\frac{1}{2}$ **(b)** $\frac{2}{3}$ **(c)** $\frac{1}{3}$ **(d)** $\frac{1}{6}$

6. A die is thrown once. The probability of getting a factor of 6 is: 1

(a) $\frac{2}{3}$ **(b)** $\frac{1}{2}$ **(c)** $\frac{1}{3}$ **(d)** $\frac{5}{6}$

7. A coin is tossed two times. The probability of getting no head is: 1

(a) 0 **(b)** $\frac{3}{4}$ **(c)** $\frac{1}{2}$ **(d)** $\frac{1}{4}$

8. A bag contains 2 white, 8 black balls. One ball is drawn at random. The probability that it is white is: 1

(a) $\frac{1}{5}$ **(b)** $\frac{3}{10}$ **(c)** $\frac{1}{10}$ **(d)** $\frac{4}{5}$

9. 15 cards numbered 1 to 15 are placed in a box, mixed well and one is drawn at random. The probability that the number drawn is divisible by both 2 and 3 is: 1

(a) $\frac{2}{15}$ **(b)** $\frac{1}{15}$ **(c)** $\frac{1}{5}$ **(d)** $\frac{13}{15}$

10. The probability of an event E is $\frac{2}{9}$. The probability of the event 'not E' is: 1

(a) $\frac{5}{18}$ **(b)** $\frac{2}{9}$ **(c)** $\frac{7}{9}$ **(d)** $\frac{11}{9}$

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. A coin is tossed two times. Find the probability of getting at least one head. 2

12. A bag contains 2 red, 6 green balls. One ball is drawn at random. Find the probability that it is red. 2

13. A die is thrown once. Find the probability of getting a composite number. 2

14. A coin is tossed two times. Find the probability of getting no head. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. In a lottery there are 5 prizes and 45 blanks. A ticket is drawn at random. Find the probability of getting a prize. 3

16. A lot of 50 ball pens contains 5 defective pens. One pen is drawn at random from the lot. Find the probability that the pen is good (not defective). 3

17. A class has 50 students, of whom 28 are girls. A student is selected at random to be the monitor. Find the probability that a boy is selected. 3

OR

In a lottery there are 12 prizes and 28 blanks. A ticket is drawn at random. Find the probability of getting a prize. 3

18. A lot of 80 ball pens contains 16 defective pens. One pen is drawn at random from the lot. Find the probability that the pen is good (not defective). 3

OR

A class has 36 students, of whom 20 are girls. A student is selected at random to be the monitor. Find the probability that a boy is selected. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. A bag contains 2 red, 3 blue, 5 green balls. One ball is drawn at random. Find the probability that the ball is: 5
(i) blue,
(ii) not red,
(iii) green or red (in this order).

OR

A box contains 15 cards numbered 1 to 15. One card is drawn at random. Find the probability that the number is: 5
(i) divisible by both 2 and 3,
(ii) greater than 12,
(iii) a number less than 16 (in this order).

20. A bag contains 6 red, 4 blue, 10 green balls. One ball is drawn at random. Find the probability that the ball is: 5
(i) red,
(ii) not green,
(iii) black (in this order).

OR

A box contains 18 cards numbered 1 to 18. One card is drawn at random. Find the probability that the number is: 5
(i) a number divisible by 5,
(ii) a perfect square,
(iii) a number less than 10 (in this order).

END OF QUESTION PAPER — Total: 40 Marks

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